

CO2-PART III

NAME: SRI JANANI B

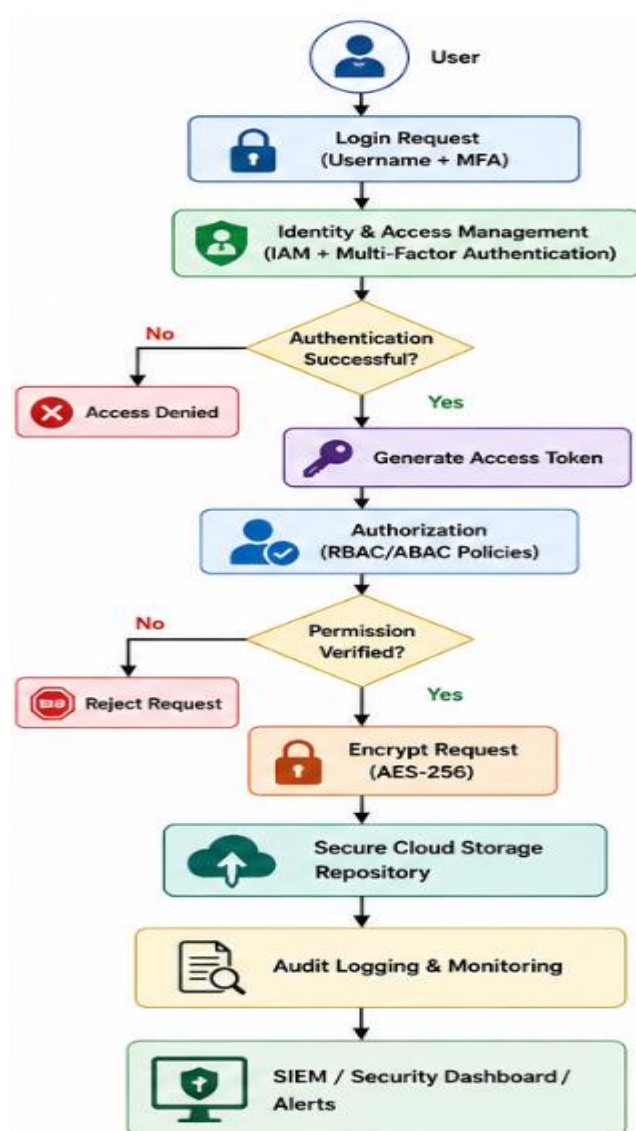
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QUESTION:

Design a secure cloud storage access workflow with encryption, identity management, auditing, and architecture diagrams.

ANSWER:

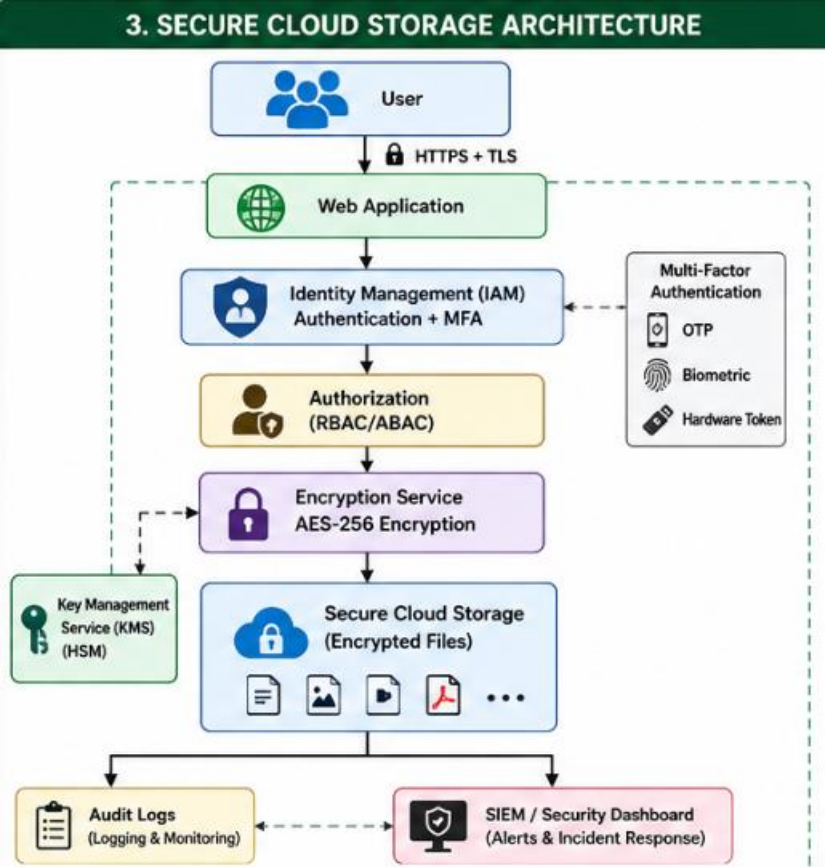
SECURE WORKFLOW MAPPING



THREAT MODELLING

Threat (STRIDE)	Description	Possible Attack	Mitigation
S Spoofing	Fake user identity	Credential theft	MFA, Strong Password Policy, IAM
T Tampering	File modification	Data corruption	AES Encryption, Hash Verification
R Repudiation	User denies action	No accountability	Immutable Audit Logs
I Information Disclosure	Data leakage	Unauthorized access	Encryption at Rest & TLS in Transit
D Denial of Service	Service unavailable	Flooding attacks	Rate Limiting, Auto Scaling, WAF
E Elevation of Privilege	Unauthorized admin access	Privilege escalation	Least Privilege, RBAC, Continuous Monitoring

SECURE CLOUD STORAGE ARCHITECTURE



Security Features Used

Security Mechanism	Purpose
Multi-Factor Authentication (MFA)	Prevents unauthorized logins
Identity and Access Management (IAM)	Manages user identities and permissions
Role-Based Access Control (RBAC)	Restricts access based on user roles
AES-256 Encryption	Protects stored data
TLS 1.3	Secures data during transmission
Key Management Service (KMS)	Protects encryption keys

Audit Logging	Ensures accountability and traceability
Security Information and Event Management (SIEM)	Monitors security events and detects threats
Least Privilege Principle	Grants only the minimum required permissions
Automatic Alerts	Detects suspicious activities in real time